Numerical Analysis

Chapter 3: Numerical Linear Algebra

Department of Mathematics & Statistics
Missouri University of Science & Technology

Textbook: Classical and Modern Numerical Analysis:
Theory, Methods and Practice
A. S. Ackleh, E. J. Allen, R. B. Kearfott and P. Seshaiyer
CRC Press, Boca Raton, FL, 2010

Outline

1. Introduction
2. Direct methods
3. Iterative methods
4. Convergence analysis

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2 Direct methods

3 Iterative methods

4 Convergence analysis

What's the target problem?

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- Iterative methods: How to construct a convergent iteration? How to obtain a faster convergence?
- .....

Review: linear algebra

Definition (Matrix)

An $m \times n$ matrix A is a rectangular array of elements arranged in m rows and n columns:

$$A = [a_{ij}]_{i=1,j=1}^{m,n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Review: linear algebra

Definition (Transpose)

The transpose of the $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ is a $n \times m$ matrix obtained from A by interchanging the rows and columns of A:

$$A^T = [a_{ji}]_{j=1,i=1}^{n,m} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \cdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

Review: linear algebra

Definition (Conjugate)

The conjugate of a matrix A is a matrix obtained from A by replacing each element a_{ij} by its conjugate $\overline{a}_{ij}$:

$$\overline{A} = [\overline{a}_{ij}]_{i=1,j=1}^{m,n} = \begin{pmatrix} \overline{a}_{11} & \overline{a}_{12} & \cdots & \overline{a}_{1n} \\ \overline{a}_{21} & \overline{a}_{22} & \cdots & \overline{a}_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ \overline{a}_{m1} & \overline{a}_{m2} & \cdots & \overline{a}_{mn} \end{pmatrix}.$$

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Definition (Adjoint)

The adjoint (conjugate transpose) of a matrix A is $A^H = \overline{A}^T$.

Review: linear algebra

- Example 1: Compute the transpose, conjugate, and adjoint of

$$A = \begin{pmatrix} 3 & 2-i \\ 4+3i & -5+2i \end{pmatrix}.$$

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Review: linear algebra

Definition (Equality)

Two $m \times n$ matrices $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and $B = [b_{ij}]_{i=1,j=1}^{m,n}$ are said to be equal if all corresponding elements are equal. That is, $a_{ij} = b_{ij}$ for each i and j.

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Definition (Multiplication by a number)

The product of an $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ by a number α is defined as $\alpha A = [\alpha a_{ij}]_{i=1,j=1}^{m,n}$.

Review: linear algebra

Definition (Addition/Subtraction)

The sum/difference of two $m \times n$ matrices $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and $B = [b_{ij}]_{i=1,j=1}^{m,n}$ are defined as $A \pm B = [a_{ij} \pm b_{ij}]_{i=1,j=1}^{m,n}$.

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Definition (Matrix multiplication)

The product of an $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and an $n \times r$ matrix $B = [b_{ij}]_{i=1,j=1}^{n,r}$ is an $m \times r$ matrix $C = [\alpha c_{ij}]_{i=1,j=1}^{m,r}$ where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Review: linear algebra

Properties of matrices:

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- In general, $AB \neq BA$.
- $(AB)^T = B^T A^T$, $(AB)^H = B^H A^H$.

Review: linear algebra

- Example 2: Compute the sum and multiplication of the matrices

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix}, \; B = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}.$$

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- Sum:

$$A + B = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 1 & -1 \\ 4 & 0 & 2 \end{pmatrix}.$$

Review: linear algebra

- Multiplication:

$$AB = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}.$$

Review: linear algebra

- Multiplication:

$$
\begin{aligned}
AB &= \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}. \\
&= \begin{pmatrix} 2-2+2 & 1+2-1 & -1+0+1 \\ 0+2-2 & 0-2+1 & 0+0-1 \\ 4+1+2 & 2-1-1 & -2+0+1 \end{pmatrix}.
\end{aligned}
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&= \begin{pmatrix} 2 & 2 & 0 \\ 0 & -1 & -1 \\ 7 & 0 & -1 \end{pmatrix}.
\end{aligned}
$$

- $BA = ?$ Try it by yourself!

Review: linear algebra

Definition (Minor)

The minor associated with each element a_{ij} of a square matrix A is the determinant of the matrix obtained by deleting the ith row and the jth column of A.

Review: linear algebra

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The minor associated with each element a_{ij} of a square matrix A is the determinant of the matrix obtained by deleting the ith row and the jth column of A.

Definition (Cofactor)

The cofactor associated with each element a_{ij} of a square matrix A is $(-1)^{i+j} M_{ij}$.

Review: linear algebra

Definition (Determinant)

The determinant of a $n \times n$ matrix A is defined as

$$det(A) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \quad \text{for a fixed } j$$

or

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Remark

$$det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Review: linear algebra

Remark

$$det(AB) = det(A)det(B).$$

Review: linear algebra

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Remark

For a lower triangular matrix

$$A = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ a_{n1} & a_{n2} & 0 & a_{nn} \end{pmatrix},$$

we also have $det(A) = \prod_{i=1}^{n} a_{ii}$.

Review: linear algebra

Remark

Similarly, for an upper triangular matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix},$$

we have $det(A) = \prod_{i=1}^{n} a_{ii}$.

Review: linear algebra

- Example 3: Compute the determinant of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 1 & -2 & -7 \end{pmatrix}.$$

Review: linear algebra

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$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 1 & -2 & -7 \end{pmatrix}.$$

- By the definition,

$$\begin{aligned} det(A) = & 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 8 \\ -2 & -7 \end{vmatrix} + 2 \cdot (-1)^{1+2} \begin{vmatrix} -2 & 8 \\ 1 & -7 \end{vmatrix} \\ & +1 \cdot (-1)^{1+3} \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \end{aligned}$$

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Review: linear algebra

Definition (Identity matrix)

The identity matrix I is a matrix whose elements are 1 on the diagonal and 0 everywhere else:

$$I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Review: linear algebra

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Remark

For any square matrix A,

$$AI = IA = A.$$

Review: linear algebra

Definition (Inverse)

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Remark

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Review: linear algebra

Remark

- $(AB)^{-1} = B^{-1}A^{-1}$.

Review: linear algebra

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- $(AB)^{-1} = B^{-1}A^{-1}$.
- $(A+B)^{-1} \neq A^{-1} + B^{-1}$.

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Remark

The inverse matrix of an upper triangular matrix is an upper triangular matrix. Similarly, the inverse matrix of a lower triangle matrix is a lower triangular matrix.

Review: linear algebra

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  (3)Addition of any multiple of one row to another row.
- We will use R_i to represent the ith row.

Review: linear algebra

- Example 4: Find the inverse of

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- First,

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right).$$

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- Example 4: Find the inverse of

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$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right).$$

- Adding $-3R_1$ to R_2 and adding $-2R_1$ to R_3:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 0 & 4 & 5 & -2 & 0 & 1 \end{array}\right).$$

Review: linear algebra

- Multiply R_2 by $\frac{1}{2}$:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 4 & 5 & -2 & 0 & 1 \end{array}\right).$$

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- Adding R_2 to R_1 and adding $-4R_2$ to R_3:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -5 & 4 & -2 & 1 \end{array}\right).$$

Review: linear algebra

- Multiply R_3 by $-\frac{1}{5}$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Review: linear algebra

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- Adding $-\frac{5}{2}R_3$ to R_2 and adding $-\frac{3}{2}R_3$ to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{10} & -\frac{1}{10} & \frac{3}{10} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Review: linear algebra

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- Adding $-\frac{5}{2}R_3$ to R_2 and adding $-\frac{3}{2}R_3$ to R_1:

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- Hence

$$A^{-1} = \left(\begin{array}{ccc} \frac{7}{10} & -\frac{1}{10} & \frac{3}{10} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Review: linear algebra

Remark

A row vector $\overrightarrow{x} = (x_1, \cdots, x_n)$ *is a* $1 \times n$ *matrix. Similarly, a column vector*

$$\overrightarrow{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (x_1, \cdots, x_n)^T$$

is a $n \times 1$ *matrix.*

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Remark

Let $\mathbb{R}^n$ *denote the* $n-$*dimensional space of* $n-$*tuples of real numbers. Let* $\mathbb{C}^n$ *denote the* $n-$*dimensional space of* $n-$*tuples of complex numbers. That is, if* $\overrightarrow{x} \in \mathbb{C}^n$*, then* $\overrightarrow{x} = (x_1, \cdots, x_n)^T$ *where* $x_i \in \mathbb{C}$ *for* $i = 1, \cdots, n$*.*

Review: linear algebra

Definition (Linearly independence/dependence)

The vectors $\vec{x}^{(1)}, \vec{x}^{(2)}, \cdots, \vec{x}^{(n)}$ are said to be linear independent if

$$\sum_{i=1}^{n} k_i \vec{x}^{(i)} = 0$$

implies $k_i = 0$ $(i = 1, 2, \cdots, n)$. Otherwise, $\vec{x}^{(i)}$ $(i = 1, 2, \cdots, n)$ are said to be linearly dependent.

Review: linear algebra

- Example 5: Determine whether the vectors

$$\vec{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \vec{x}^{(2)} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

  are linearly independent or linearly dependent.

Review: linear algebra

- Example 5: Determine whether the vectors

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Review: linear algebra

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$$k_1 \vec{x}^{(1)} + k_2 \vec{x}^{(2)} = 0.$$

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$$\begin{aligned} k_1 - 4k_2 &= 0, \\ 2k_1 + k_2 &= 0. \end{aligned}$$

Review: linear algebra

- Hence

$$2k_1 - 8k_2 = 0,$$
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Review: linear algebra

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$$2k_1 - 8k_2 = 0,$$
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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

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Review: linear algebra

Theorem (I)

Let A be an $n \times n$ matrix. Then the following are equivalent:

- *A is nonsingular/invertible.*

Review: linear algebra

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Review: linear algebra

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- *The columns (and rows) of A are linearly independent.*

Review: linear algebra

Definition (Eigenvalue/eigenvector)

A value λ and a non-trivial vector $\overrightarrow{x}$ satisfying

$$A\overrightarrow{x} = \lambda\overrightarrow{x}$$

are called an eigenvalue and eigenvector of the matrix A respectively.

Review: linear algebra

Remark

- *First,*

$$\begin{aligned} & A\vec{x} = \lambda\vec{x} \\ \Rightarrow \quad & A\vec{x} - \lambda\vec{x} = 0 \\ \Rightarrow \quad & (A - \lambda I)\vec{x} = 0 \end{aligned}$$

where I is the identity matrix.

Review: linear algebra

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  where I is the identity matrix.

- *Since the eigenvector $\vec{x}$ needs to be non-trivial, then by Theorem (I), we obtain the characteristic equation of A:*

$$det(A - \lambda I) = 0.$$

Review: linear algebra

- Example 7: Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

Review: linear algebra

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- We compute the eigenvalues first:

$$\begin{aligned} 0 &= det(A - \lambda I) \\ &= \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} \\ &= (3-\lambda)(-2-\lambda) + 4 \\ &= -6 - 3\lambda + 2\lambda + \lambda^2 + 4 \\ &= \lambda^2 - \lambda - 2 \\ &= (\lambda - 2)(\lambda + 1). \end{aligned}$$

Review: linear algebra

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- Eigenvalues: $\lambda_1 = 2$ and $\lambda_2 = -1$.

Review: linear algebra

- Now we compute the eigenvector corresponding to $\lambda_1 = 2$:

$$
\begin{aligned}
& (A - \lambda I)\vec{x} = 0 \\
\Rightarrow \quad & (A - \lambda_1 I)\vec{\xi}^{(1)} = 0 \\
\Rightarrow \quad & \begin{pmatrix} 3-2 & -1 \\ 4 & -2-2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
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- Hence $\xi_1 - \xi_2 = 0$.

- Pick $\xi_2 = 1$, then $\xi_1 = 1$. Hence

$$\vec{\xi}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Review: linear algebra

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

$$
\begin{aligned}
& (A - \lambda I)\vec{x} = 0 \\
\Rightarrow \quad & (A - \lambda_2 I)\vec{\xi}^{(2)} = 0 \\
\Rightarrow \quad & \begin{pmatrix} 3-(-1) & -1 \\ 4 & -2-(-1) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
\Rightarrow \quad & \begin{pmatrix} 4 & -1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

Review: linear algebra

- Hence $4\xi_1 - \xi_2 = 0$.

Review: linear algebra

- Hence $4\xi_1 - \xi_2 = 0$.
- Pick $\xi_1 = 1$, then $\xi_2 = 4$. Hence

$$\vec{\xi}^{(2)} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Review: linear algebra

Definition (spectral radius)

$\rho(A) = \max_{1 \le i \le n} |\lambda_i|$, where $\{\lambda_i\}_{i=1}^n$ is the set of eigenvalues of A, is called the spectral radius of A.

Review: linear algebra

Definition (spectral radius)

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Definition (positive definite matrix)

If $\overrightarrow{x}^T A \overrightarrow{x} > 0$ for any $\overrightarrow{x} \neq 0$, then A is said to be positive definite.

Review: linear algebra

Theorem (II)

If A is symmetric positive definite or Hermitian positive definite, then its eigenvalues are real and positive.

Review: linear algebra

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If system $B\overrightarrow{x} = \overrightarrow{d}$ is obtained from system $A\overrightarrow{x} = \overrightarrow{b}$ by a finite sequence of elementary row operations, then the two systems have the same solutions.

Review: linear algebra

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Theorem (III)

A $n \times n$ matrix is non-singular if and only if no eigenvalues are 0.

Outline

1 Introduction

2 Direct methods

3 Iterative methods

4 Convergence analysis

Basic idea of Gaussian elimination

- Target problem: given a nonsingular matrix $A = [a_{ij}]_{i,j=1}^{n}$ and a column vector $\overrightarrow{b} = [b_i]_{i=1}^{n}$, find $\overrightarrow{x} = [x_i]_{i=1}^{n}$ such that $A\overrightarrow{x} = \overrightarrow{b}$, that is,

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\
\cdots &, \\
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- Basic idea: use the three elementary row operations to reduce the matrix A to be an upper triangular matrix U; then use back substitution to solve the system.

Basic idea of Gaussian elimination

Recall

Theorem

Let A be an $n \times n$ matrix. Then the following are equivalent:

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- *The columns (and rows) of A are linearly independent.*

Simple examples of Gaussian elimination

- Example 1: Use Gaussian elimination to solve the system of equations

$$
\begin{aligned}
x_1 - 2x_2 + 3x_3 &= 7, \\
-x_1 + x_2 - 2x_3 &= -5, \\
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  (1)Interchanging of two rows.

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- Elementary row operations:

  (1)Interchanging of two rows.

  (2)Multiplication of a row by a nonzero scalar.

  (3)Addition of any multiple of one row to another row.

Simple examples of Gaussian elimination

- Matrix formulation of the system:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ -1 & 1 & -2 & -5 \\ 2 & -1 & -1 & 4 \end{array}\right).$$

Simple examples of Gaussian elimination

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- Goal: change the matrix to be an upper triangular matrix by using the row operations. That is, change the elements in the lower triangle part of the matrix to be 0.

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- Goal: change the matrix to be an upper triangular matrix by using the row operations. That is, change the elements in the lower triangle part of the matrix to be 0.
- We will use R_i to represent the ith row.

Simple examples of Gaussian elimination

- Let $m_{21} = \frac{-1}{1}$ and add $-m_{21}R_1 = R_1$ to R_2:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & -1 & 1 & 2 \\ 2 & -1 & -1 & 4 \end{array}\right).$$

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- Let $m_{31} = \frac{2}{1}$ and add $-m_{31}R_1 = -2R_1$ to R_3:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & -1 & 1 & 2 \\ 0 & 3 & -7 & -10 \end{array}\right).$$

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- Let $m_{32} = \frac{3}{-1}$ and add $-m_{32}R_2 = 3R_2$ to R_3:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & -4 & -4 \end{array}\right).$$

Simple examples of Gaussian elimination

- Hence

$$
\begin{aligned}
x_1 - 2x_2 + 3x_3 &= 7, \\
-x_2 + x_3 &= 2, \\
-4x_3 &= -4.
\end{aligned}
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Simple examples of Gaussian elimination

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\begin{aligned}
x_1 - 2x_2 + 3x_3 &= 7, \\
-x_2 + x_3 &= 2, \\
-4x_3 &= -4.
\end{aligned}
$$

- back substitution (back solving):

$$
\begin{aligned}
&-4x_3 = -4 \Rightarrow x_3 = 1, \\
&-x_2 + x_3 = 2 \Rightarrow -x_2 + 1 = 2 \Rightarrow x_2 = -1, \\
&x_1 - 2x_2 + 3x_3 = 7 \Rightarrow x_1 - 2 \cdot (-1) + 3 \cdot 1 = 7 \Rightarrow x_1 = 2.
\end{aligned}
$$

Simple examples of Gaussian elimination

- Example 2: Use Gaussian elimination to solve the system of equations

$$
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x_1 - 2x_2 + 3x_3 &= 7, \\
-x_1 + 2x_2 - 2x_3 &= -5, \\
2x_1 - x_2 - x_3 &= 4.
\end{aligned}
$$

Simple examples of Gaussian elimination

- Example 2: Use Gaussian elimination to solve the system of equations

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- Matrix formulation of the system:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ -1 & 2 & -2 & -5 \\ 2 & -1 & -1 & 4 \end{array}\right).$$

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$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 0 & 1 & 2 \\ 0 & 3 & -7 & -10 \end{array}\right).$$

- Interchange R_2 and R_3:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 3 & -7 & -10 \\ 0 & 0 & 1 & 2 \end{array}\right).$$

Simple examples of Gaussian elimination

- Hence

$$
\begin{aligned}
x_1 - 2x_2 + 3x_3 &= 7, \\
3x_2 - 7x_3 &= -10, \\
x_3 &= 2.
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$$

Simple examples of Gaussian elimination

- Hence

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 7, \\ 3x_2 - 7x_3 &= -10, \\ x_3 &= 2. \end{aligned}$$

- back substitution (back solving):

$$\begin{aligned} & x_3 = 2, \\ & 3x_2 - 7x_3 = -10 \Rightarrow 3x_2 - 7 \cdot 2 = -10 \Rightarrow x_2 = \frac{4}{3}, \\ & x_1 - 2x_2 + 3x_3 = 7 \Rightarrow x_1 - 2 \cdot \frac{4}{3} + 3 \cdot 2 = 7 \Rightarrow x_1 = \frac{11}{3}. \end{aligned}$$

Gaussian elimination procedure

- Step 0 (preparation): Let $A^{(1)} = A$ and $\overrightarrow{b}^{(1)} = \overrightarrow{b}$. So the system becomes $A^{(1)}\overrightarrow{x} = \overrightarrow{b}^{(1)}$ with

$$A^{(1)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}^{(1)} & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

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$$\overrightarrow{b}^{(1)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

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- Step 1: Assume $a_{11}^{(1)} \neq 0$.

Gaussian elimination procedure

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
- Let $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.

Gaussian elimination procedure

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
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- Add $-m_{i1} R_1$ to R_i $(2 \leq i \leq n)$ in $A^{(1)}$ and $\overrightarrow{b}^{(1)}$ to obtain $A^{(2)}\overrightarrow{x} = \overrightarrow{b}^{(2)}$ with

$$A^{(2)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} \end{pmatrix}, \quad \overrightarrow{b}^{(2)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(2)} \end{pmatrix}.$$

Gaussian elimination procedure

- More details for understanding step 1:

Gaussian elimination procedure

- More details for understanding step 1:
- Question 1: How can we guarantee the assumption ($a_{11}^{(1)} \neq 0$) is reasonable?

Gaussian elimination procedure

- More details for understanding step 1:
- Question 1: How can we guarantee the assumption ($a_{11}^{(1)} \neq 0$) is reasonable?
- By using the nonsingularity of the matrix A.

Gaussian elimination procedure

Recall

Definition (Determinant)

The determinant of a $n \times n$ matrix A is defined as

$$det(A) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \quad \text{for a fixed } j$$

or

$$det(A) = \sum_{j=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \quad \text{for a fixed } i.$$

Gaussian elimination procedure

Recall

Theorem

Let A be an $n \times n$ matrix. Then the following are equivalent:

- *A is nonsingular/invertible.*

Gaussian elimination procedure

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Gaussian elimination procedure

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Theorem

Let A be an $n \times n$ matrix. Then the following are equivalent:

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Gaussian elimination procedure

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Gaussian elimination procedure

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Let A be an $n \times n$ matrix. Then the following are equivalent:

- *A is nonsingular/invertible.*
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- *$\overrightarrow{x} = 0$ is the unique solution of $A\overrightarrow{x} = 0$.*
- *$A\overrightarrow{x} = \overrightarrow{b}$ has a unique solution.*
- *The columns (and rows) of A are linearly independent.*

Gaussian elimination procedure

- If the elements of the first column of $A^{(1)}$ are all zeros ($a_{i1}^{(1)} = 0$ for $1 \leq i \leq n$), then

$$det(A) = det\left(A^{(1)}\right) = 0.$$

Gaussian elimination procedure

- If the elements of the first column of $A^{(1)}$ are all zeros ($a_{i1}^{(1)} = 0$ for $1 \leq i \leq n$), then

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- This contradicts the nonsingularity of the matrix A based on the above theorem.

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- Hence at least one element of the first column of $A^{(1)}$ is nonzero. That is, there exists a s such that $a_{s1}^{(1)} \neq 0$ and $1 \leq s \leq n$.

Gaussian elimination procedure

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- If $a_{11}^{(1)} = 0$, then the first row of $A^{(1)}$ can be interchanged with the sth row such that the new $a_{11}^{(1)}$ is nonzero.

Gaussian elimination procedure

- If the elements of the first column of $A^{(1)}$ are all zeros ($a_{i1}^{(1)} = 0$ for $1 \leq i \leq n$), then

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- This contradicts the nonsingularity of the matrix A based on the above theorem.
- Hence at least one element of the first column of $A^{(1)}$ is nonzero. That is, there exists a s such that $a_{s1}^{(1)} \neq 0$ and $1 \leq s \leq n$.
- If $a_{11}^{(1)} = 0$, then the first row of $A^{(1)}$ can be interchanged with the sth row such that the new $a_{11}^{(1)}$ is nonzero.
- Therefore, $a_{11}^{(1)} \neq 0$ is a reasonable practical assumption.

Gaussian elimination procedure

- Question 2: How do we obtain $A^{(2)}$ and $b^{(2)}$?

Gaussian elimination procedure

- Question 2: How do we obtain $A^{(2)}$ and $b^{(2)}$?
- Since $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$, then

$$
\begin{aligned}
-m_{i1}R_1 &= -m_{i1}\left(\begin{array}{cccc|c} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} & b_1^{(1)} \end{array}\right) \\
&= \left(\begin{array}{cccc|c} -a_{i1}^{(1)} & -\frac{a_{i1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} & \cdots & -\frac{a_{i1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} & -\frac{a_{i1}^{(1)}b_1^{(1)}}{a_{11}^{(1)}} \end{array}\right)
\end{aligned}
$$

Gaussian elimination procedure

- Hence

$$
\begin{aligned}
& -m_{i1}R_1 + R_i \\
= & \left(\begin{array}{cccc|c} -a_{i1}^{(1)} & -\frac{a_{i1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} & \cdots & -\frac{a_{i1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} & -\frac{a_{i1}^{(1)}b_1^{(1)}}{a_{11}^{(1)}} \end{array} \right) \\
& + \left(\begin{array}{cccc|c} a_{i1}^{(1)} & a_{i2}^{(1)} & \cdots & a_{in}^{(1)} & b_i^{(1)} \end{array} \right) \\
= & \left(\begin{array}{cccc|c} 0 & -\frac{a_{i1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{i2}^{(1)} & \cdots & -\frac{a_{i1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{in}^{(1)} & -\frac{a_{i1}^{(1)}b_1^{(1)}}{a_{11}^{(1)}} + b_i^{(1)} \end{array} \right)
\end{aligned}
$$

Gaussian elimination procedure

- Adding $-m_{i1}R_1$ to R_i $(2 \leq i \leq n)$ in $A^{(1)}$ and $b^{(1)}$, we get

$$\begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & -\frac{a_{21}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{22}^{(1)} & \cdots & -\frac{a_{21}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & -\frac{a_{n1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{n2}^{(1)} & \cdots & -\frac{a_{n1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{nn}^{(1)} \end{pmatrix}$$

$$\triangleq \quad A^{(2)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} \end{pmatrix}$$

Gaussian elimination procedure

- And

$$\begin{pmatrix} b_1^{(1)} \\ -\frac{a_{21}^{(1)} b_1^{(1)}}{a_{11}^{(1)}} + b_2^{(1)} \\ \vdots \\ -\frac{a_{n1}^{(1)} b_1^{(1)}}{a_{11}^{(1)}} + b_n^{(1)} \end{pmatrix} \triangleq \overrightarrow{b}(2) = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(2)} \end{pmatrix}.$$

Gaussian elimination procedure

Recall

- Step 0 (preparation): Let $A^{(1)} = A$ and $\overrightarrow{b}^{(1)} = \overrightarrow{b}$. So the system becomes $A^{(1)}\overrightarrow{x} = \overrightarrow{b}^{(1)}$ with

$$A^{(1)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}^{(1)} & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

$$\overrightarrow{b}^{(1)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

Gaussian elimination procedure

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.

Gaussian elimination procedure

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
- Let $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.

Gaussian elimination procedure

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
- Let $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.
- Add $-m_{i1}R_1$ to R_i $(2 \leq i \leq n)$ in $A^{(1)}$ and $\overrightarrow{b}^{(1)}$ to obtain $A^{(2)}\overrightarrow{x} = \overrightarrow{b}^{(2)}$ with

$$A^{(2)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} \end{pmatrix}, \quad \overrightarrow{b}^{(2)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(2)} \end{pmatrix}.$$

Gaussian elimination procedure

- Step 2: Assume $a_{22}^{(2)} \neq 0$. This assumption is reasonable since the nonsingularity of the matrix A guarantees that there exists a s such that $a_{s2}^{(2)} \neq 0$ $(2 \leq s \leq n)$ (Homework).

Gaussian elimination procedure

- Step 2: Assume $a_{22}^{(2)} \neq 0$. This assumption is reasonable since the nonsingularity of the matrix A guarantees that there exists a s such that $a_{s2}^{(2)} \neq 0$ $(2 \leq s \leq n)$ (Homework).
- Let $m_{i2} = \frac{a_{i2}^{(2)}}{a_{22}^{(2)}}$ $(3 \leq i \leq n)$.

Gaussian elimination procedure

- Step 2: Assume $a_{22}^{(2)} \neq 0$. This assumption is reasonable since the nonsingularity of the matrix A guarantees that there exists a s such that $a_{s2}^{(2)} \neq 0$ $(2 \leq s \leq n)$ (Homework).
- Let $m_{i2} = \frac{a_{i2}^{(2)}}{a_{22}^{(2)}}$ $(3 \leq i \leq n)$.
- Add $-m_{i2}R_2$ to R_i $(3 \leq i \leq n)$ in $A^{(2)}$ and $\overrightarrow{b}^{(2)}$ to obtain $A^{(3)}\overrightarrow{x} = \overrightarrow{b}^{(3)}$ with

$$A^{(3)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2n}^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3n}^{(3)} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & a_{n3}^{(3)} & \cdots & a_{nn}^{(3)} \end{pmatrix}, \quad \overrightarrow{b}^{(3)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ b_3^{(3)} \\ \vdots \\ b_n^{(3)} \end{pmatrix}.$$

Gaussian elimination procedure

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.

Gaussian elimination procedure

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.
- Let $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ $(k+1 \leq i \leq n)$.

Gaussian elimination procedure

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.
- Let $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ $(k+1 \leq i \leq n)$.
- Add $-m_{ik}R_k$ to R_i $(k+1 \leq i \leq n)$ in $A^{(k)}$ and $\overrightarrow{b}^k$ to obtain $A^{(k+1)}\overrightarrow{x} = \overrightarrow{b}^{(k+1)}$ with

$$A^{(k+1)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1k}^{(1)} & a_{1,k+1}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2k}^{(2)} & a_{2,k+1}^{(2)} & \cdots & a_{2n}^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3k}^{(3)} & a_{3,k+1}^{(3)} & \cdots & a_{3n}^{(3)} \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \ddots & a_{kk}^{(k)} & a_{k,k+1}^{(k)} & \cdots & a_{kn}^{(k)} \\ 0 & 0 & \cdots & \cdots & 0 & a_{k+1,k+1}^{(k+1)} & \cdots & a_{k+1,n}^{(k+1)} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & a_{n,k+1}^{(k+1)} & \cdots & a_{nn}^{(k+1)} \end{pmatrix},$$

Gaussian elimination procedure

- and

$$\overrightarrow{b}^{(k+1)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ b_3^{(3)} \\ \vdots \\ b_k^{(k)} \\ b_{k+1}^{(k+1)} \\ \vdots \\ b_n^{(k+1)} \end{pmatrix}.$$

Gaussian elimination procedure

- Finally, we obtain $A^{(n)}\overrightarrow{x} = \overrightarrow{b}^{(n)}$ with

$$A^{(n)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn}^{(n)} \end{pmatrix} \triangleq \begin{pmatrix} a_{11}^{(n)} & a_{12}^{(n)} & \cdots & a_{1n}^{(n)} \\ 0 & a_{22}^{(n)} & \cdots & a_{2n}^{(n)} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn}^{(n)} \end{pmatrix}$$

$$\overrightarrow{b}^{(n)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(n)} \end{pmatrix} \triangleq \begin{pmatrix} b_1^{(n)} \\ b_2^{(n)} \\ \vdots \\ b_n^{(n)} \end{pmatrix}.$$

Gaussian elimination procedure

- That is,

$$\begin{pmatrix} a_{11}^{(n)} & a_{12}^{(n)} & \cdots & a_{1n}^{(n)} \\ 0 & a_{22}^{(n)} & \cdots & a_{2n}^{(n)} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn}^{(n)} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1^{(n)} \\ b_2^{(n)} \\ \vdots \\ b_n^{(n)} \end{pmatrix}.$$

Gaussian elimination procedure

- Then

$$
\begin{aligned}
&a_{11}^{(n)} x_1 + a_{12}^{(n)} x_2 + \cdots + a_{1n}^{(n)} x_n = b_1^{(n)} \\
&a_{22}^{(n)} x_2 + a_{23}^{(n)} x_3 + \cdots + a_{2n}^{(n)} x_n = b_2^{(n)} \\
&\cdots , \\
&a_{n-2,n-2}^{(n)} x_{n-2} + a_{n-2,n-1}^{(n)} x_{n-1} + a_{n-2,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{n-1,n-1}^{(n)} x_{n-1} + a_{n-1,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{nn}^{(n)} x_n = b_n^{(n)}
\end{aligned}
$$

Gaussian elimination procedure

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&\cdots , \\
&a_{n-2,n-2}^{(n)} x_{n-2} + a_{n-2,n-1}^{(n)} x_{n-1} + a_{n-2,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{n-1,n-1}^{(n)} x_{n-1} + a_{n-1,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{nn}^{(n)} x_n = b_n^{(n)}
\end{aligned}
$$

- Back substitution:

$$
x_n = \frac{b_n^{(n)}}{a_{nn}^{(n)}}, \; x_{n-1} = \frac{b_{n-1}^{(n)} - a_{n-1,n}^{(n)} x_n}{a_{n-1,n-1}^{(n)}},
$$

Gaussian elimination procedure

$$x_{n-2} = \frac{b_{n-2}^{(n)} - \left[a_{n-2,n-1}^{(n)} x_{n-1} + a_{n-2,n}^{(n)} x_n\right]}{a_{n-1,n-1}^{(n)}},$$

Gaussian elimination procedure

$$
\begin{aligned}
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&\vdots \\
x_2 &= \frac{b_2^{(n)} - \left[a_{23}^{(n)} x_3 + \cdots + a_{2n}^{(n)} x_n\right]}{a_{22}^{(n)}},
\end{aligned}
$$

Gaussian elimination procedure

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&\vdots \\
x_2 &= \frac{b_2^{(n)} - \left[a_{23}^{(n)} x_3 + \cdots + a_{2n}^{(n)} x_n\right]}{a_{22}^{(n)}}, \\
x_1 &= \frac{b_1^{(n)} - \left[a_{12}^{(n)} x_2 + \cdots + a_{1n}^{(n)} x_n\right]}{a_{11}^{(n)}}.
\end{aligned}
$$

Gaussian elimination procedure

- In fact,

$$
\begin{aligned}
&a_{11}^{(n)} x_1 + a_{12}^{(n)} x_2 + \cdots + a_{1n}^{(n)} x_n = b_1^{(n)} \\
&a_{22}^{(n)} x_2 + a_{23}^{(n)} x_3 + \cdots + a_{2n}^{(n)} x_n = b_2^{(n)} \\
&\cdots , \\
&a_{n-2,n-2}^{(n)} x_{n-2} + a_{n-2,n-1}^{(n)} x_{n-1} + a_{n-2,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{n-1,n-1}^{(n)} x_{n-1} + a_{n-1,n}^{(n)} x_n = b_{n-1}^{(n)} \\
&a_{nn}^{(n)} x_n = b_n^{(n)}
\end{aligned}
$$

Gaussian elimination procedure

- In fact,

$$
\begin{aligned}
&a_{11}^{(n)}x_1 + a_{12}^{(n)}x_2 + \cdots + a_{1n}^{(n)}x_n = b_1^{(n)}\\
&a_{22}^{(n)}x_2 + a_{23}^{(n)}x_3 + \cdots + a_{2n}^{(n)}x_n = b_2^{(n)}\\
&\cdots,\\
&a_{n-2,n-2}^{(n)}x_{n-2} + a_{n-2,n-1}^{(n)}x_{n-1} + a_{n-2,n}^{(n)}x_n = b_{n-1}^{(n)}\\
&a_{n-1,n-1}^{(n)}x_{n-1} + a_{n-1,n}^{(n)}x_n = b_{n-1}^{(n)}\\
&a_{nn}^{(n)}x_n = b_n^{(n)}
\end{aligned}
$$

can be rewritten as

$$
\begin{aligned}
&a_{nn}^{(n)}x_n = b_n^{(n)},\\
&a_{kk}^{(n)}x_k + \sum_{j=k+1}^{n} a_{kj}^{(n)}x_j = b_k^{(n)},\ \ k = n-1, n-2, \cdots, 1.
\end{aligned}
$$

Gaussian elimination procedure

- Conclusion of back substitution:

$$x_n = \frac{b_n^{(n)}}{a_{nn}^{(n)}},$$

Gaussian elimination procedure

- Conclusion of back substitution:

$$
\begin{aligned}
x_n &= \frac{b_n^{(n)}}{a_{nn}^{(n)}}, \\
x_k &= \frac{b_k^{(n)} - \sum_{j=k+1}^{n} a_{kj}^{(n)} x_j}{a_{kk}^{(n)}}, \quad k = n-1, n-2, \cdots, 1.
\end{aligned}
$$

Algorithm of Gaussian elimination

Key observations:

- Input and preparation.

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- Loop for $A^{(k+1)}$, $\overrightarrow{b}^{(k)}$ $(k = 1, 2, \cdots, n-1)$: treating $A^{(k)}$ and $\overrightarrow{b}^{(k)}$ row by row to generate $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$.

Algorithm of Gaussian elimination

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- Input and preparation.
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- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.

Algorithm of Gaussian elimination

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- Input and preparation.
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- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- Loop for treating the ith $(k+1 \leq i \leq n)$ row of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$ to obtain the ith row of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$: $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ and add $-m_{ik} R_k$ to R_i in $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.

Algorithm of Gaussian elimination

Key observations:

- Input and preparation.
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- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- Loop for treating the ith $(k+1 \leq i \leq n)$ row of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$ to obtain the ith row of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$: $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ and add $-m_{ik}R_k$ to R_i in $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- Loop for treating the jth $(j = k, \cdots, n)$ element of the ith row of $A^{(k)}$: add $-m_{ik}R_k$ to R_i element by element, that is, add $-m_{ik}a_{kj}^{(k)}$ to $a_{ij}^{(k)}$ and add $-m_{ik}b_k^{(k)}$ to $b_i^{(k)}$.

Algorithm of Gaussian elimination

Version 1 for forward phase:

1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.

Algorithm of Gaussian elimination

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1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. Initial preparation: $A^{(1)} = A$ and $\overrightarrow{b}^{(1)} = \overrightarrow{b}$.

Algorithm of Gaussian elimination

Version 1 for forward phase:

1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. Initial preparation: $A^{(1)} = A$ and $\overrightarrow{b}^{(1)} = \overrightarrow{b}$.
3. *for* $k = 1, 2, \cdots, n-1$:

   $A^{(k+1)} = A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = \overrightarrow{b}^{(k)}$.

   for $i = k+1, \cdots, n$:

   - (1) $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$.
   - (2) *for* $j = k, \cdots, n$:

     $$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)};$$

     END
   - (3) $b_i^{(k+1)} = b_i^{(k)} - m_{ik} b_k^{(k)}$.

   END

END

Algorithm of Gaussian elimination

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   $A^{(k+1)} = A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = \overrightarrow{b}^{(k)}$.

   for $i = k+1, \cdots, n$:

   - (1) $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$.
   - (2) *for* $j = k, \cdots, n$:

     $$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)};$$

     END
   - (3) $b_i^{(k+1)} = b_i^{(k)} - m_{ik} b_k^{(k)}$.

   END

   END
4. Output $A^{(n)}$ and $\overrightarrow{b}^{(n)}$.

Algorithm of Gaussian elimination

More observations:

- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.

Algorithm of Gaussian elimination

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- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- In order to obtain $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$, the kth row of $A^{(k)}$ and $\overrightarrow{b}^{(k+1)}$ is used to treat the rows below it.

Algorithm of Gaussian elimination

More observations:

- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- In order to obtain $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$, the kth row of $A^{(k)}$ and $\overrightarrow{b}^{(k+1)}$ is used to treat the rows below it.
- So we can directly overwrite the $A^{(k)}$ and $\overrightarrow{b}^{(k)}$ to obtain $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$.

Algorithm of Gaussian elimination

More observations:

- The first k rows of $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$ are the same as those of $A^{(k)}$ and $\overrightarrow{b}^{(k)}$.
- In order to obtain $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$, the kth row of $A^{(k)}$ and $\overrightarrow{b}^{(k+1)}$ is used to treat the rows below it.
- So we can directly overwrite the $A^{(k)}$ and $\overrightarrow{b}^{(k)}$ to obtain $A^{(k+1)}$ and $\overrightarrow{b}^{(k+1)}$.
- Hence, we only need to allocate the memory for one matrix A and use it to handle all of $A^{(k)}$ $(k = 1, \cdots, n)$.

Algorithm of Gaussian elimination

Final algorithm for forward phase:

1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.

Algorithm of Gaussian elimination

Final algorithm for forward phase:

1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. *FOR* $k = 1, 2, \cdots, n-1$:

   FOR $i = k+1, \cdots, n$:

   - (1) $m_{ik} = \frac{a_{ik}}{a_{kk}}$.
   - (2) *FOR* $j = k, \cdots, n$:

     $a_{ij} = a_{ij} - m_{ik} a_{kj}$;

     END
   - (3) $b_i = b_i - m_{ik} b_k$.

   END

END

Algorithm of Gaussian elimination

Final algorithm for forward phase:

1. Input the $n \times n$ matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. *FOR* $k = 1, 2, \cdots, n-1$:
   FOR $i = k+1, \cdots, n$:
   - (1) $m_{ik} = \frac{a_{ik}}{a_{kk}}$.
   - (2) *FOR* $j = k, \cdots, n$:
     $a_{ij} = a_{ij} - m_{ik} a_{kj}$;
     END
   - (3) $b_i = b_i - m_{ik} b_k$.

   END

   END
3. Output A and $\overrightarrow{b}$.

Algorithm of Gaussian elimination

Recall

- Conclusion of back substitution:

$$
\begin{aligned}
x_n &= \frac{b_n^{(n)}}{a_{nn}^{(n)}}, \\
x_k &= \frac{b_k^{(n)} - \sum_{j=k+1}^{n} a_{kj}^{(n)} x_j}{a_{kk}^{(n)}}, \quad k = n-1, n-2, \cdots, 1.
\end{aligned}
$$

Algorithm of Gaussian elimination

Algorithm of back substitution phase:

1. Input the $n \times n$ upper triangular matrix A and $n \times 1$ vector $\overrightarrow{b}$.

Algorithm of Gaussian elimination

Algorithm of back substitution phase:

1. Input the $n \times n$ upper triangular matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. $x_n = \frac{b_n}{a_{nn}}$.

Algorithm of Gaussian elimination

Algorithm of back substitution phase:

1. Input the $n \times n$ upper triangular matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. $x_n = \frac{b_n}{a_{nn}}$.
3. *FOR* $k = n-1, n-2, \cdots, 1$:

$$x_k = \frac{b_k - \sum_{j=k+1}^{n} a_{kj} x_j}{a_{kk}}.$$

Algorithm of Gaussian elimination

Algorithm of back substitution phase:

1. Input the $n \times n$ upper triangular matrix A and $n \times 1$ vector $\overrightarrow{b}$.
2. $x_n = \frac{b_n}{a_{nn}}$.
3. *FOR* $k = n-1, n-2, \cdots, 1$:
$$x_k = \frac{b_k - \sum_{j=k+1}^{n} a_{kj} x_j}{a_{kk}}.$$
4. Output the solution $\overrightarrow{x} = [x_i]_{i=1}^{n}$.

Algorithm of Gaussian elimination

Remark

The forward phase of Gauss elimination requires $\frac{1}{3}n^3 + O(n^2)$ *multiplications and divisions. The back solving Gauss elimination requires* $\frac{n^2+n}{2}$ *multiplications and divisions. (Homework)*

Practice in Matlab for Gaussian elimination

- Example 3: Solve the following system $A\vec{x} = \vec{b}$ by using Gaussian elimination:

$$\begin{pmatrix} 7 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 8 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 9 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 10 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 11 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 12 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 13 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{pmatrix}$$

Practice in Matlab for Gaussian elimination

- Example 3: Solve the following system $A\vec{x} = \vec{b}$ by using Gaussian elimination:

$$\begin{pmatrix} 7 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 8 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 9 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 10 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 11 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 12 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 13 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{pmatrix}$$

- Let's code for this algorithm together!
- Open your Matlab!

Practice in Matlab for Gaussian elimination

- Numerical results:

$$\overrightarrow{x} = \begin{pmatrix} -0.0990 \\ 0.0580 \\ 0.1758 \\ 0.2673 \\ 0.3406 \\ 0.4006 \\ 0.4505 \end{pmatrix}$$

Practice in Matlab for Gaussian elimination

- Verification:

$$A\vec{x} - \vec{b} = 10^{-14} \times \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -0.1776 \\ -0.0888 \\ 0.0888 \end{pmatrix}.$$

LU decomposition

- Basic idea: Each elementary row operation on a matrix A is equivalent to multiply A by an invertible matrix.

LU decomposition

- Basic idea: Each elementary row operation on a matrix A is equivalent to multiply A by an invertible matrix.
- What's that invertible matrix?

LU decomposition

- Basic idea: Each elementary row operation on a matrix A is equivalent to multiply A by an invertible matrix.
- What's that invertible matrix?
- Let's start from the $n \times n$ identity matrix

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}_{n \times n}.$$

LU decomposition

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.

LU decomposition

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
- Let $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.

LU decomposition

Recall

- Step 1: Assume $a_{11}^{(1)} \neq 0$.
- Let $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.
- Add $-m_{i1}R_1$ to R_i $(2 \leq i \leq n)$ in $A^{(1)}$ and $\overrightarrow{b}^{(1)}$ to obtain $A^{(2)}\overrightarrow{x} = \overrightarrow{b}^{(2)}$ with

$$A^{(2)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} \end{pmatrix}, \quad \overrightarrow{b}^{(2)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(2)} \end{pmatrix}.$$

LU decomposition

Recall

- Adding $-m_{i1}R_1$ to R_i $(2 \leq i \leq n)$ in $A^{(1)}$ and $b^{(1)}$, we get

$$\begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & -\frac{a_{21}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{22}^{(1)} & \cdots & -\frac{a_{21}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & -\frac{a_{n1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{n2}^{(1)} & \cdots & -\frac{a_{n1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{nn}^{(1)} \end{pmatrix}$$

$$\triangleq \quad A^{(2)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} \end{pmatrix}$$

LU decomposition

- Add $-m_{i1}R_1$ to R_i $(2 \leq i \leq n)$ in I_n to obtain

$$M^{(1)} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ -m_{n1} & 0 & \cdots & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & & & \\ \vdots & & I_{n-1} & \\ -m_{n1} & & & \end{pmatrix}.$$

LU decomposition

- Let's compute $M^{(1)}A^{(1)}$ with $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.

LU decomposition

- Let's compute $M^{(1)}A^{(1)}$ with $m_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}}$ $(2 \leq i \leq n)$.
- We get

$$M^{(1)}A^{(1)} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -m_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ -m_{n1} & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1}^{(1)} & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & -\frac{a_{21}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{22}^{(1)} & \cdots & -\frac{a_{21}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{2n}^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & -\frac{a_{n1}^{(1)}a_{12}^{(1)}}{a_{11}^{(1)}} + a_{n2}^{(1)} & \cdots & -\frac{a_{n1}^{(1)}a_{1n}^{(1)}}{a_{11}^{(1)}} + a_{nn}^{(1)} \end{pmatrix}.$$

LU decomposition

- Hence $A^{(2)} = M^{(1)} A^{(1)}$.

LU decomposition

- Hence $A^{(2)} = M^{(1)} A^{(1)}$.
- Similarly, $\overrightarrow{b}^{(2)} = M^{(1)} \overrightarrow{b}^{(1)}$.

LU decomposition

Recall:

Remark

$$det(AB) = det(A)det(B).$$

LU decomposition

Recall:

Remark

$$det(AB) = det(A)det(B).$$

Remark

Similarly, for a lower triangular matrix

$$A = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ a_{n1} & a_{n2} & 0 & a_{nn} \end{pmatrix},$$

we also have $det(A) = \prod_{i=1}^{n} a_{ii}$.

LU decomposition

- Hence $det\left(M^{(1)}\right) = 1$.

LU decomposition

- Hence $det\left(M^{(1)}\right) = 1$.
- And

$$\begin{aligned} det\left(A^{(2)}\right) &= det\left(M^{(1)}A^{(1)}\right) \\ &= det\left(M^{(1)}\right) det\left(A^{(1)}\right) \\ &= det\left(A^{(1)}\right). \end{aligned}$$

LU decomposition

Recall

- Step 2: Assume $a_{22}^{(2)} \neq 0$.

LU decomposition

Recall

- Step 2: Assume $a_{22}^{(2)} \neq 0$.
- Let $m_{i2} = \frac{a_{i2}^{(2)}}{a_{22}^{(2)}}$ $(3 \leq i \leq n)$.

LU decomposition

Recall

- Step 2: Assume $a_{22}^{(2)} \neq 0$.
- Let $m_{i2} = \frac{a_{i2}^{(2)}}{a_{22}^{(2)}}$ $(3 \leq i \leq n)$.
- Add $-m_{i2}R_2$ to R_i $(3 \leq i \leq n)$ in $A^{(2)}$ and $\overrightarrow{b}^{(2)}$ to obtain $A^{(3)}\overrightarrow{x} = \overrightarrow{b}^{(3)}$ with

$$A^{(3)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2n}^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3n}^{(3)} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & a_{n3}^{(3)} & \cdots & a_{nn}^{(3)} \end{pmatrix}, \quad \overrightarrow{b}^{(3)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ b_3^{(3)} \\ \vdots \\ b_n^{(3)} \end{pmatrix}.$$

LU decomposition

Recall

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.

LU decomposition

Recall

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.
- Let $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ $(k+1 \leq i \leq n)$.

LU decomposition

Recall

- Step 3 to $n-1$: Repeat the above process. Consider the kth step where $3 \leq k \leq n-1$. Assume $a_{kk}^{(k)} \neq 0$.
- Let $m_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$ $(k+1 \leq i \leq n)$.
- Add $-m_{ik}R_k$ to R_i $(k+1 \leq i \leq n)$ in $A^{(k)}$ and $\overrightarrow{b}^k$ to obtain $A^{(k+1)}\overrightarrow{x} = \overrightarrow{b}^{(k+1)}$ with

$$A^{(k+1)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1k}^{(1)} & a_{1,k+1}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2k}^{(2)} & a_{2,k+1}^{(2)} & \cdots & a_{2n}^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3k}^{(3)} & a_{3,k+1}^{(3)} & \cdots & a_{3n}^{(3)} \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \ddots & a_{kk}^{(k)} & a_{k,k+1}^{(k)} & \cdots & a_{kn}^{(k)} \\ 0 & 0 & \cdots & \cdots & 0 & a_{k+1,k+1}^{(k+1)} & \cdots & a_{k+1,n}^{(k+1)} \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & a_{n,k+1}^{(k+1)} & \cdots & a_{nn}^{(k+1)} \end{pmatrix},$$

LU decomposition

Recall

- and

$$\overrightarrow{b}^{(k+1)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ b_3^{(3)} \\ \vdots \\ b_k^{(k)} \\ b_{k+1}^{(k+1)} \\ \vdots \\ b_n^{(k+1)} \end{pmatrix}.$$

LU decomposition

- Add $-m_{ik}R_k$ to R_i $(k+1 \leq i \leq n)$ in I_n to obtain

$$M^{(k)} = \begin{pmatrix} I_{k-1} & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -m_{k+1,k} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & -m_{n,k} & 0 & \cdots & 1 \end{pmatrix}$$

LU decomposition

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$$M^{(k)} = \begin{pmatrix} I_{k-1} & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -m_{k+1,k} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & -m_{n,k} & 0 & \cdots & 1 \end{pmatrix}$$

$$= \begin{pmatrix} I_{k-1} & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & -m_{k+1,k} & & & \\ \vdots & \vdots & & I_{n-k} & \\ 0 & -m_{n,k} & & & \end{pmatrix}.$$

LU decomposition

- Similarly, $A^{(k+1)} = M^{(k)} A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = M^{(k)} \overrightarrow{b}^{(k)}$.

LU decomposition

- Similarly, $A^{(k+1)} = M^{(k)} A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = M^{(k)} \overrightarrow{b}^{(k)}$.
- Then we obtain $A^{(n)} \overrightarrow{x} = \overrightarrow{b}^{(n)}$ with

$$\begin{aligned} A^{(n)} &= M^{(n-1)} A^{(n-1)} \\ &= M^{(n-1)} M^{(n-2)} A^{(n-2)} \\ &= M^{(n-1)} M^{(n-2)} \dots M^{(1)} A^{(1)} \end{aligned}$$

LU decomposition

- Similarly, $A^{(k+1)} = M^{(k)} A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = M^{(k)} \overrightarrow{b}^{(k)}$.
- Then we obtain $A^{(n)} \overrightarrow{x} = \overrightarrow{b}^{(n)}$ with

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and

$$\overrightarrow{b}^{(n)} = M^{(n-1)} M^{(n-2)} \dots M^{(1)} \overrightarrow{b}^{(1)}.$$

LU decomposition

- Similarly, $A^{(k+1)} = M^{(k)} A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = M^{(k)} \overrightarrow{b}^{(k)}$.
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and

$$\overrightarrow{b}^{(n)} = M^{(n-1)} M^{(n-2)} \dots M^{(1)} \overrightarrow{b}^{(1)}.$$

- Then

$$A = A^{(1)} = \left(M^{(n-1)} M^{(n-2)} \dots M^{(1)} \right)^{-1} A^{(n)}$$

LU decomposition

- Similarly, $A^{(k+1)} = M^{(k)} A^{(k)}$ and $\overrightarrow{b}^{(k+1)} = M^{(k)} \overrightarrow{b}^{(k)}$.
- Then we obtain $A^{(n)} \overrightarrow{x} = \overrightarrow{b}^{(n)}$ with

$$\begin{aligned} A^{(n)} &= M^{(n-1)} A^{(n-1)} \\ &= M^{(n-1)} M^{(n-2)} A^{(n-2)} \\ &= M^{(n-1)} M^{(n-2)} \dots M^{(1)} A^{(1)} \end{aligned}$$

and

$$\overrightarrow{b}^{(n)} = M^{(n-1)} M^{(n-2)} \dots M^{(1)} \overrightarrow{b}^{(1)}.$$

- Then

$$A = A^{(1)} = \left(M^{(n-1)} M^{(n-2)} \dots M^{(1)} \right)^{-1} A^{(n)}$$

- Let $L = \left(M^{(n-1)} M^{(n-2)} \dots M^{(1)} \right)^{-1}$ and $U = A^{(n)}$. Then

$$A = LU.$$

LU decomposition

Recall

- Finally, we obtain $A^{(n)}\overrightarrow{x} = \overrightarrow{b}^{(n)}$ with

$$A^{(n)} = \begin{pmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn}^{(n)} \end{pmatrix} \triangleq \begin{pmatrix} a_{11}^{(n)} & a_{12}^{(n)} & \cdots & a_{1n}^{(n)} \\ 0 & a_{22}^{(n)} & \cdots & a_{2n}^{(n)} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn}^{(n)} \end{pmatrix}$$

$$\overrightarrow{b}^{(n)} = \begin{pmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_n^{(n)} \end{pmatrix} \triangleq \begin{pmatrix} b_1^{(n)} \\ b_2^{(n)} \\ \vdots \\ b_n^{(n)} \end{pmatrix}.$$

LU decomposition

- Hence $U = A^{(n)}$ is an upper triangular matrix.

LU decomposition

- Hence $U = A^{(n)}$ is an upper triangular matrix.
- Since $M^{(k)}$ $(k = 1, \cdots, n-1)$ are all of lower triangle matrix, then $M^{(n-1)}M^{(n-2)}\cdots M^{(1)}$ is a lower triangle matrix.

LU decomposition

- Hence $U = A^{(n)}$ is an upper triangular matrix.
- Since $M^{(k)}$ $(k = 1, \cdots, n-1)$ are all of lower triangle matrix, then $M^{(n-1)}M^{(n-2)}\cdots M^{(1)}$ is a lower triangle matrix.
- Hence $L = \left(M^{(n-1)}M^{(n-2)}\cdots M^{(1)}\right)^{-1}$ is a lower triangular matrix. Actually

$$L = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ m_{21} & 1 & 0 & \cdots & 0 \\ m_{31} & m_{32} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{n,n-1} & 1 \end{pmatrix}.$$

Topics: Independent study problems

- Pivoting Gaussian elimination (section 3.3.1.4): interchange the rows and (or) columns of the matrix in order to choose the leading terms $a_{kk}^{(k)}$.

Topics: Independent study problems

- Pivoting Gaussian elimination (section 3.3.1.4): interchange the rows and (or) columns of the matrix in order to choose the leading terms $a_{kk}^{(k)}$.
- Cholesky decomposition (section 3.3.2): LU decomposition for symmetric positive definite matrices.

Topics: Independent study problems

- Pivoting Gaussian elimination (section 3.3.1.4): interchange the rows and (or) columns of the matrix in order to choose the leading terms $a_{kk}^{(k)}$.
- Cholesky decomposition (section 3.3.2): LU decomposition for symmetric positive definite matrices.
- QR decomposition (section 3.3.8): $A = QR$ where R is an upper triangular matrix and Q is an orthogonal matrix ($Q^T Q = I$); orthogonal decomposition, Householder transformation, Givens rotation, Gram-Schmidt procedure.

Topics: Independent study problems

- Pivoting Gaussian elimination (section 3.3.1.4): interchange the rows and (or) columns of the matrix in order to choose the leading terms $a_{kk}^{(k)}$.
- Cholesky decomposition (section 3.3.2): LU decomposition for symmetric positive definite matrices.
- QR decomposition (section 3.3.8): $A = QR$ where R is an upper triangular matrix and Q is an orthogonal matrix ($Q^T Q = I$); orthogonal decomposition, Householder transformation, Givens rotation, Gram-Schmidt procedure.
- Singular value decomposition (SVD) (section 3.5): $A = U\Sigma V^T$ where Σ is a diagonal matrix and U and V are orthogonal matrices; the diagonal elements of Σ are the singular values.

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)
- LU decomposition for tridiagonal matrices (section 3.3.3)

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)
- LU decomposition for tridiagonal matrices (section 3.3.3)
- LU decomposition for block tridiagonal matrices (section 3.3.4)

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)
- LU decomposition for tridiagonal matrices (section 3.3.3)
- LU decomposition for block tridiagonal matrices (section 3.3.4)
- Roundoff error and conditioning in Gaussian elimination (section 3.3.5; see section 3.2 for norms)

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)
- LU decomposition for tridiagonal matrices (section 3.3.3)
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- Roundoff error and conditioning in Gaussian elimination (section 3.3.5; see section 3.2 for norms)
- Iterative refinement (section 3.3.6)

Topics: Independent study problems

- LU decomposition for symmetric positive definite matrices (section 3.3.2)
- LU decomposition for tridiagonal matrices (section 3.3.3)
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- Roundoff error and conditioning in Gaussian elimination (section 3.3.5; see section 3.2 for norms)
- Iterative refinement (section 3.3.6)
- Interval bounds (section 3.3.7)

Outline

1 Introduction

2 Direct methods

3 Iterative methods

4 Convergence analysis

Introduction

- Observation 1: For many real world applications, we need to solve a huge system $A\overrightarrow{x} = \overrightarrow{b}$. That is, A is a nonsingular matrix with a huge dimension.

Introduction

- Observation 1: For many real world applications, we need to solve a huge system $A\overrightarrow{x} = \overrightarrow{b}$. That is, A is a nonsingular matrix with a huge dimension.
- Observation 2: It is difficult to solve a huge system $A\overrightarrow{x} = \overrightarrow{b}$ by using the direct methods introduced above.

Introduction

- Observation 1: For many real world applications, we need to solve a huge system $A\overrightarrow{x} = \overrightarrow{b}$. That is, A is a nonsingular matrix with a huge dimension.
- Observation 2: It is difficult to solve a huge system $A\overrightarrow{x} = \overrightarrow{b}$ by using the direct methods introduced above.
- Conclusion: iterative methods are more practical to solve a huge system.

Introduction

- Basic idea: Split A as $A = M - N$ where M is a nonsingular matrix.

Introduction

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- Cauchy convergence: Given any $\varepsilon > 0$, there exists a N such that $\left\| \vec{x}^{(m)} - \vec{x}^{(n)} \right\| < \varepsilon$ for all $m, n \geq N$.
- Given a small enough tolerance $\varepsilon > 0$. Then the iteration can be stopped when

$$\left\| \vec{x}^{(k+1)} - \vec{x}^{(k)} \right\| \leq \varepsilon.$$

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- Consider

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad \overrightarrow{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

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- Define

$$D = diag\left(a_{11}, \cdots, a_{nn}\right) = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix},$$

Introduction

- Define

$$L = \begin{pmatrix} 0 & \cdots & \cdots & 0 \\ a_{21} & \ddots & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & \cdots & a_{n,n-1} & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & a_{12} & \cdots & a_{1n} \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & a_{n-1,n} \\ 0 & \cdots & \cdots & 0 \end{pmatrix}.$$

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- Then $A = D + L + U$.

The Jacobi method

- Choose $M = D$ so that it's easy to compute M^{-1}. Then $A = M - N$ leads to

$$N = M - A = D - (D + L + U) = -(L + U)$$

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- If $a_{ii} \neq 0$ $(1 \leq i \leq n)$, then $det(M) = \prod_{i=1}^{n} a_{ii} \neq 0$. Hence M is nonsingular/invertible.
- Actually,

$$M^{-1} = D^{-1} = diag\left(\frac{1}{a_{11}}, \cdots, \frac{1}{a_{nn}}\right) = \begin{pmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{pmatrix}.$$

Again, the easy computation of M^{-1} is one major motivation of the Jacobi method.

The Jacobi method

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$$\begin{aligned} \overrightarrow{x}^{(k+1)} &= -D^{-1}(L+U)\overrightarrow{x}^{(k)} + D^{-1}\overrightarrow{b}, \ k = 0, 1, 2, \cdots \\ &= D^{-1}\overrightarrow{b} - D^{-1}L\overrightarrow{x}^{(k)} - D^{-1}U\overrightarrow{x}^{(k)}. \end{aligned}$$

The Jacobi method

- Then

$$\begin{pmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \\ \vdots \\ x_n^{(k+1)} \end{pmatrix} = \begin{pmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

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- Hence

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left\{ b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right\},$$

$$i = 1, \cdots, n, \;\; k = 0, 1, 2, \cdots$$

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$$i = 1, \cdots, n, \;\; k = 0, 1, 2, \cdots$$

- Observation: When we compute $x_i^{(k+1)}$, $x_j^{(k+1)}$ are already obtained for $j = 1, \cdots, i-1$. But the Jacobi method still uses $x_j^{(k)}$ $(j = 1, \cdots, i-1)$, which are not he newest information.

The Gauss-Seidel (GS) method

- The Gauss-Seidel (GS) method (or successive relaxation method):

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left\{ b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \right\},$$

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Remark

In fact, the Gauss-Seidel method is an iterative method with $M = L + D$ *and* $N = -U$. *That is*

$$\vec{x}^{(k+1)} = M^{-1} N \vec{x}^{(k)} + M^{-1} \vec{b}, \ \ k = 0, 1, 2, \cdots$$
$$\Rightarrow \quad \vec{x}^{(k+1)} = -(L + D)^{-1} U \vec{x}^{(k)} + (L + D)^{-1} \vec{b}, \ \ k = 0, 1, 2, \cdots$$

The Gauss-Seidel (GS) method

Proof:

- Recall the matrix formulation of the Jacobi method:

$$\overrightarrow{x}^{(k+1)} = D^{-1}\overrightarrow{b} - D^{-1}L\overrightarrow{x}^{(k)} - D^{-1}U\overrightarrow{x}^{(k)}, \ k = 0, 1, 2, \cdots$$

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The successive overrelaxation (SOR) method

- Basic idea: add a relaxation factor σ into the Gauss-Seidel method (successive relaxation method).

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$$x_i^{(k+1)} = (1-\sigma)x_i^{(k)} + \frac{\sigma}{a_{ii}}\left\{ b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij}x_j^{(k)} \right\},$$

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$$i = 1, \cdots, n, \;\; k = 0, 1, 2, \cdots$$

- σ is called an overrelaxation factor if $\sigma > 1$ and an underrelaxation factor if $\sigma < 1$.

The successive overrelaxation (SOR) method

Remark

In fact, the SOR method is an iterative method with $M = L + \frac{1}{\sigma}D$ *and* $N = -\left(U + D - \frac{1}{\sigma}D\right)$. *That is*

$$\vec{x}^{(k+1)} = M^{-1}N\vec{x}^{(k)} + M^{-1}\vec{b}, \ k = 0, 1, 2, \cdots$$

$$\Rightarrow \quad \vec{x}^{(k+1)} = \left(L + \frac{1}{\sigma}D\right)^{-1}\left(U + D - \frac{1}{\sigma}D\right)\vec{x}^{(k)} + \left(L + \frac{1}{\sigma}D\right)^{-1}\vec{b}, \ k = 0, 1, 2, \cdots$$

Homework: Prove the above remark.

The successive overrelaxation (SOR) method

Remark

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Homework: Prove the above remark.

Remark

For certain classes of matrices and certain $\sigma \in (1, 2)$, *the SOR method converges faster than the GS method.*

Practice in Matlab for Jacobi, GS and SOR methods

- Example 1: Solve the following system $A\overrightarrow{x} = \overrightarrow{b}$ by using Jacobi, GS and SOR methods:

$$\begin{pmatrix} 7 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 8 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 9 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 10 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 11 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 12 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 13 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{pmatrix}$$

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- Let's code for these algorithms together!
- Open your Matlab!

Practice in Matlab for Jacobi, GS and SOR methods

Recall the numerical results of example 3 for Gaussian elimination:

- Numerical results:

$$\vec{x} = \begin{pmatrix} -0.0990 \\ 0.0580 \\ 0.1758 \\ 0.2673 \\ 0.3406 \\ 0.4006 \\ 0.4505 \end{pmatrix}$$

Practice in Matlab for Jacobi, GS and SOR methods

Recall the numerical results of example 3 for Gaussian elimination:

- Verification:

$$A\vec{x} - \vec{b} = 10^{-14} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -0.1776 \\ -0.0888 \\ 0.0888 \end{pmatrix}.$$

Practice in Matlab for Jacobi, GS and SOR methods

- Numerical results with tolerance $\varepsilon = 10^{-10}$ and trivial initial guess for example 1:

$$\vec{x}_n = \begin{pmatrix} -0.0990 \\ 0.0580 \\ 0.1758 \\ 0.2673 \\ 0.3406 \\ 0.4006 \\ 0.4505 \end{pmatrix}$$

where $n = 50$ for the Jacobi method, $n = 13$ for GS method, and $n = 44$ for SOR method with $\sigma = 1.5$.

Practice in Matlab for Jacobi, GS and SOR methods

- Verification for Jacobi:

$$A\overrightarrow{x} - \overrightarrow{b} = 10^{-9} \begin{pmatrix} -0.1692 \\ -0.1733 \\ -0.1766 \\ -0.1793 \\ -0.1816 \\ -0.1835 \\ -0.1852 \end{pmatrix}.$$

Practice in Matlab for Jacobi, GS and SOR methods

- Verification for GS:

$$A\overrightarrow{x} - \overrightarrow{b} = 10^{-9} \begin{pmatrix} 0.1006 \\ 0.0556 \\ 0.0156 \\ -0.0064 \\ -0.0124 \\ -0.0084 \\ 0.0000 \end{pmatrix}.$$

Practice in Matlab for Jacobi, GS and SOR methods

- Verification for OR with $\sigma = 1.5$:

$$A\overrightarrow{x} - \overrightarrow{b} = 10^{-9} \begin{pmatrix} 0.1014 \\ 0.1615 \\ 0.1626 \\ 0.1276 \\ 0.0791 \\ 0.0325 \\ -0.0041 \end{pmatrix}.$$

Topics: Independent study problems

- Multigrid method
- Preconditioned conjugate gradient method (PCG): symmetric positive definite matrix.
- Generalized minimum residual method (GMRES)
- Krylov subspace methods (section 3.4.12)
- Interval Gauss-Seidel method (section 3.4.5)
- Block SOR method (section 3.4.9)
- Iterative methods for matrix inversion (section 3.4.11)
- .....

Outline

1 Introduction

2 Direct methods

3 Iterative methods

4 Convergence analysis

Vector norms

Definition (Norms for vectors)

For a vector $\overrightarrow{x} = (x_1, \cdots, x_n)^T \in \mathbb{C}^n$, important norms are

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- l_p norm: $\left\|\vec{x}\right\|_p = \left(\sum\limits_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$.

Vector norms

Definition (Convergence in norm)

A sequence of vectors $\{\vec{x}^k\}_{k=1}^{\infty}$ is said to converge to a vector $\vec{x}$ if and only if

$$\lim_{k\to\infty} \left\| \vec{x}^k - \vec{x} \right\| = 0$$

for some norm $\|\cdot\|$.

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Example

Frobenius norm: $\|A\|_E = \left(\sum_{i,j=1}^{n} |a_{ij}|^2 \right)^{\frac{1}{2}}$.

Matrix norms

Definition (Induced matrix norm)

Given a vector norm $\|\cdot\|$, the induced (or natural) matrix norm is defined as

$$\|A\| = \sup_{\vec{x} \neq 0} \frac{\|A\vec{x}\|}{\|\vec{x}\|}.$$

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$$\Rightarrow \quad \|A\vec{x}\| \leq \|A\| \, \|\vec{x}\| .$$

Matrix norms

Remark

The above definition is equivalent to

$$\|A\| = \sup_{\|\vec{y}\|=1} \left\|A\vec{y}\right\|.$$

Proof:

- $\left\|A\frac{\vec{x}}{\|\vec{x}\|}\right\| \|\vec{x}\| = \|A\vec{x}\| \frac{1}{\|\vec{x}\|} \|\vec{x}\| = \|A\vec{x}\|.$

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- Then $\frac{\|A\vec{x}\|}{\|\vec{x}\|} = \left\|A\frac{\vec{x}}{\|\vec{x}\|}\right\|.$

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- Then $\frac{\|A\vec{x}\|}{\|\vec{x}\|} = \left\|A\frac{\vec{x}}{\|\vec{x}\|}\right\|.$
- Let $\vec{y} = \frac{\vec{x}}{\|\vec{x}\|}$. Then

$$\|A\| = \sup_{\vec{x}\neq 0} \frac{\|A\vec{x}\|}{\|\vec{x}\|} = \sup_{\vec{x}\neq 0} \left\|A\frac{\vec{x}}{\|\vec{x}\|}\right\| = \sup_{\|\vec{y}\|=1} \left\|A\vec{y}\right\|.$$

Matrix norms

Remark

We shall use the matrix norms $\|\cdot\|_\infty$, $\|\cdot\|_1$, and $\|\cdot\|_2$ induced by the corresponding vector norms. That is,

$$\|A\|_\infty = \sup_{\vec{x}\neq 0} \frac{\|A\vec{x}\|_\infty}{\|\vec{x}\|_\infty}, \quad \|A\|_1 = \sup_{\vec{x}\neq 0} \frac{\|A\vec{x}\|_1}{\|\vec{x}\|_1}, \|A\|_2 = \sup_{\vec{x}\neq 0} \frac{\|A\vec{x}\|_2}{\|\vec{x}\|_2}.$$

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Recall

Definition (spectral radius)

$\rho(A) = \max_{1\leq i\leq n} |\lambda_i|$, where $\{\lambda_i\}_{i=1}^n$ is the set of eigenvalues of A, is called the spectral radius of A.

Matrix norms

Lemma (I)

For a square matrix $A = [a_{ij}]_{i,j=1}^{n}$,

$$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}| = \{\textit{maximum absolute row sum}\},$$

Matrix norms

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For a square matrix $A = [a_{ij}]_{i,j=1}^{n}$,

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Matrix norms

Lemma (I)

For a square matrix $A = [a_{ij}]_{i,j=1}^{n}$,

$$\begin{aligned} \|A\|_\infty &= \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}| = \{\textit{maximum absolute row sum}\}, \\ \|A\|_1 &= \max_{1\le j\le n} \sum_{i=1}^{n} |a_{ij}| = \{\textit{maximum absolute column sum}\}, \\ \|A\|_2 &= \sqrt{\rho\left(A^H A\right)}. \end{aligned}$$

Proof.

See page 98-100 of the textbook. (Independent study problem) □

Matrix norms

Lemma (II)

If $\|\cdot\|$ is any matrix norm, then

$$\rho(A) \leq \|A\| .$$

Proof.

See page 95-96 of the textbook. (Independent study problem) □

Matrix norms

Lemma (III)

For a square matrix A and any $\varepsilon > 0$, there exists an induced matrix norm $\|\cdot\|$ such that $\rho(A) + \varepsilon \geq \|A\|$.

Proof.

See page 101-102 of the textbook. (Independent study problem) □

Matrix norms

Definition (Convergence in matrix)

A series of matrices $\{A_k\}_{k=1}^{\infty}$ converges to matrix A if and only if

$$\lim_{k\to\infty} \|A_k - A\| = 0.$$

Matrix norms

Lemma (IV)

Given an $n \times n$ matrix A, the following are equivalent:

- *(a)* $\lim_{k\to\infty} A^k = 0$ *(i.e,* $\lim_{k\to\infty} \left\|A^k\right\| = 0$*).*
- *(b)* $\lim_{k\to\infty} A^k \overrightarrow{y} = 0,\ \forall \overrightarrow{y} \in \mathbb{C}^n$ *(i.e,* $\lim_{k\to\infty} \left\|A^k \overrightarrow{y}\right\| = 0$*).*
- *(c)* $\rho(A) < 1$*.*
- *(d) There exists a norm* $\|\cdot\|$ *such that* $\|A\| < 1$*.*

Matrix norms

Proof:

- We will prove that $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$.

Matrix norms

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  Since

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Matrix norms

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  then

$$\lim_{k \to \infty} \left\| A^k \overrightarrow{y} \right\| \leq \lim_{k \to \infty} \left\| A^k \right\| \left\| \overrightarrow{y} \right\| = \left\| \overrightarrow{y} \right\| \lim_{k \to \infty} \left\| A^k \right\|.$$

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- From (a) $\lim_{k \to \infty} \left\| A^k \right\| = 0$, we get
  (b) $\lim_{k \to \infty} \left\| A^k \overrightarrow{y} \right\| = 0, \ \forall \overrightarrow{y} \in \mathbb{C}^n$.

Matrix norms

Continued proof:

- $(b) \Rightarrow (c)$:
  If λ is an eigenvalue and $\overrightarrow{v}$ is the corresponding eigenvector, then $A\overrightarrow{v} = \lambda\overrightarrow{v}$.

Matrix norms

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Matrix norms

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- Hence

$$A^k\overrightarrow{v} = A^{k-1}A\overrightarrow{v} = \lambda A^{k-1}\overrightarrow{v} = \cdots = \lambda^k\overrightarrow{v}$$

- Then by $\|\lambda A\| = |\lambda| \, \|A\|$, we have

$$\lim_{k\to\infty} \left\|A^k\overrightarrow{v}\right\| = \lim_{k\to\infty} \left\|\lambda^k\overrightarrow{v}\right\| = \lim_{k\to\infty} \left|\lambda^k\right| \left\|\overrightarrow{v}\right\| = \lim_{k\to\infty} |\lambda|^k \left\|\overrightarrow{v}\right\|.$$

Matrix norms

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- Then from (b) $\lim_{k\to\infty} \left\|A^k \overrightarrow{y}\right\| = 0,\ \forall \overrightarrow{y} \in \mathbb{C}^n$, we get

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$$\lim_{k\to\infty} |\lambda|^k \left\|\overrightarrow{v}\right\| = 0.$$

- Since $\overrightarrow{v} \neq 0$, then

$$\begin{aligned} & \lim_{k\to\infty} |\lambda|^k = 0 \\ \Rightarrow \quad & |\lambda| < 1. \end{aligned}$$

Matrix norms

Continued proof:

- Then from (b) $\lim_{k\to\infty} \left\|A^k \overrightarrow{y}\right\| = 0,\ \forall \overrightarrow{y} \in \mathbb{C}^n$, we get

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- Since $\overrightarrow{v} \neq 0$, then

$$\begin{aligned} & \lim_{k\to\infty} |\lambda|^k = 0 \\ \Rightarrow \quad & |\lambda| < 1. \end{aligned}$$

- Since $\rho(A) = \max_{1\leq i\leq n} |\lambda_i|$ where $\{\lambda_i\}_{i=1}^n$ is the set of eigenvalues of A, then we get (c) $\rho(A) < 1$.

Matrix norms

Continued proof:

- $(c) \Rightarrow (d)$:
  Based on (c) $\rho(A) < 1$, we can choose a $\varepsilon < 1 - \rho(A)$.

Matrix norms

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- $(c) \Rightarrow (d)$:
  Based on (c) $\rho(A) < 1$, we can choose a $\varepsilon < 1 - \rho(A)$.
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  Based on (c) $\rho(A) < 1$, we can choose a $\varepsilon < 1 - \rho(A)$.
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- Then by Lemma (III), there exist a norm $\|\cdot\|$ such that $\|A\| \leq \rho(A) + \varepsilon$

Matrix norms

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- $(c) \Rightarrow (d)$:
  Based on (c) $\rho(A) < 1$, we can choose a $\varepsilon < 1 - \rho(A)$.
- Then $\rho(A) + \varepsilon < 1$.
- Then by Lemma (III), there exist a norm $\|\cdot\|$ such that $\|A\| \leq \rho(A) + \varepsilon$
- Hence we obtain (d) there exist a norm $\|\cdot\|$ such that $\|A\| < 1$.

Matrix norms

Continued proof:

- $(d) \Rightarrow (a)$:
  Since $\|AB\| \leq \|A\| \, \|B\|$, then $\left\|A^k\right\| \leq \|A\|^k$.

Matrix norms

Continued proof:

- $(d) \Rightarrow (a)$:
  Since $\|AB\| \leq \|A\| \, \|B\|$, then $\left\|A^k\right\| \leq \|A\|^k$.
- Also, $\|A\| \geq 0$.

Matrix norms

Continued proof:

- $(d) \Rightarrow (a)$:
  Since $\|AB\| \leq \|A\| \, \|B\|$, then $\left\|A^k\right\| \leq \|A\|^k$.
- Also, $\|A\| \geq 0$.
- Then from (d) there exist a norm $\|\cdot\|$ such that $\|A\| < 1$, we get (a) $\lim_{k \to \infty} \left\|A^k\right\| = 0$.

Convergence of the traditional iterative methods

Recall

- Basic idea: Split A as $A = M - N$ where M is a nonsingular matrix.

Convergence of the traditional iterative methods

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- Basic idea: Split A as $A = M - N$ where M is a nonsingular matrix.
- Then $A\vec{x} = \vec{b}$ becomes

$$(M - N)\vec{x} = \vec{b} \Rightarrow M\vec{x} = N\vec{x} + \vec{b} \Rightarrow \vec{x} = M^{-1}N\vec{x} + M^{-1}\vec{b}.$$

Convergence of the traditional iterative methods

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- Given $\vec{x}^{(0)} = [x_i^{(0)}]_{i=1}^n$, then by the idea of the fixed point method:

$$\vec{x}^{(k+1)} = M^{-1}N\vec{x}^{(k)} + M^{-1}\vec{b}, \ k = 0, 1, 2, \cdots$$

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  $$\vec{x}^{(k+1)} = M^{-1}N\vec{x}^{(k)} + M^{-1}\vec{b}, \ k = 0, 1, 2, \cdots$$

- Define $B = M^{-1}N$ and $\vec{c} = M^{-1}\vec{b}$, then

  $$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c}, \ k = 0, 1, 2, \cdots$$

  and

  $$\vec{x} = B\vec{x} + \vec{c}.$$

Convergence of the traditional iterative methods

Definition

The iterative method

$$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c}, \ k = 0, 1, 2, \cdots$$

is called convergent if

$$\lim_{k \to \infty} \vec{x}^{(k)} = \vec{x} = A^{-1}\vec{b}$$

for all initial values $\vec{x}^{(0)}$.

Convergence of the traditional iterative methods

Theorem (I)

Consider the iterative method

$$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c}, \ k = 0, 1, 2, \cdots$$

The following are equivalent:

- *(a) The iterative method is convergent.*

Convergence of the traditional iterative methods

Theorem (I)

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- *(b)* $\rho(B) < 1$.

Convergence of the traditional iterative methods

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Consider the iterative method

$$\overrightarrow{x}^{(k+1)} = B\overrightarrow{x}^{(k)} + \overrightarrow{c}, \ k = 0, 1, 2, \cdots$$

The following are equivalent:

- *(a) The iterative method is convergent.*
- *(b)* $\rho(B) < 1$.
- *(c) There exists a matrix norm* $\|\cdot\|$ *such that* $\|B\| < 1$.

Convergence of the traditional iterative methods

Proof:

- Since $\overrightarrow{x}^{(k)} = B\overrightarrow{x}^{(k-1)} + \overrightarrow{c}$ and $\overrightarrow{x} = B\overrightarrow{x} + \overrightarrow{c}$, then

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Proof:

- Since $\vec{x}^{(k)} = B\vec{x}^{(k-1)} + \vec{c}$ and $\vec{x} = B\vec{x} + \vec{c}$, then

$$\vec{x}^{(k)} - \vec{x} = B\left(\vec{x}^{(k-1)} - \vec{x}\right)$$

Convergence of the traditional iterative methods

Proof:

- Since $\vec{x}^{(k)} = B\vec{x}^{(k-1)} + \vec{c}$ and $\vec{x} = B\vec{x} + \vec{c}$, then

$$\begin{aligned} \vec{x}^{(k)} - \vec{x} &= B\left(\vec{x}^{(k-1)} - \vec{x}\right) \\ &= B^2\left(\vec{x}^{(k-2)} - \vec{x}\right) \end{aligned}$$

Convergence of the traditional iterative methods

Proof:

- Since $\vec{x}^{(k)} = B\vec{x}^{(k-1)} + \vec{c}$ and $\vec{x} = B\vec{x} + \vec{c}$, then

$$\begin{aligned}
\vec{x}^{(k)} - \vec{x} &= B\left(\vec{x}^{(k-1)} - \vec{x}\right) \\
&= B^2\left(\vec{x}^{(k-2)} - \vec{x}\right) \\
&= \cdots \\
&= B^k\left(\vec{x}^{(0)} - \vec{x}\right)
\end{aligned}$$

Convergence of the traditional iterative methods

Proof:

- Since $\vec{x}^{(k)} = B\vec{x}^{(k-1)} + \vec{c}$ and $\vec{x} = B\vec{x} + \vec{c}$, then

$$\begin{aligned} \vec{x}^{(k)} - \vec{x} &= B\left(\vec{x}^{(k-1)} - \vec{x}\right) \\ &= B^2\left(\vec{x}^{(k-2)} - \vec{x}\right) \\ &= \cdots \\ &= B^k\left(\vec{x}^{(0)} - \vec{x}\right) \end{aligned}$$

- Let $\vec{y} = \vec{x}^{(0)} - \vec{x}$. Then (a) is equivalent to

$$\lim_{k\to\infty} B^k\vec{y} = 0$$

for all $\vec{y}$.

Matrix norms

Continued proof: Recall

Lemma (IV)

Given an $n \times n$ matrix A, the following are equivalent:

- *(a)* $\lim_{k\to\infty} A^k = 0$ *(i.e,* $\lim_{k\to\infty} \left\|A^k\right\| = 0$*).*

Matrix norms

Continued proof: Recall

Lemma (IV)

Given an $n \times n$ matrix A, the following are equivalent:

- *(a)* $\lim_{k\to\infty} A^k = 0$ *(i.e,* $\lim_{k\to\infty} \left\|A^k\right\| = 0$*).*
- *(b)* $\lim_{k\to\infty} A^k \overrightarrow{y} = 0,\ \forall \overrightarrow{y} \in \mathbb{C}^n$ *(i.e,* $\lim_{k\to\infty} \left\|A^k \overrightarrow{y}\right\| = 0$*).*

Matrix norms

Continued proof: Recall

Lemma (IV)

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- *(c)* $\rho(A) < 1$*.*

Matrix norms

Continued proof: Recall

Lemma (IV)

Given an $n \times n$ matrix A, the following are equivalent:

- *(a)* $\lim_{k\to\infty} A^k = 0$ *(i.e,* $\lim_{k\to\infty} \left\|A^k\right\| = 0$*).*
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- *(c)* $\rho(A) < 1$*.*
- *(d) There exists a norm* $\|\cdot\|$ *such that* $\|A\| < 1$*.*

Matrix norms

Continued proof: Recall

Lemma (IV)

Given an $n \times n$ matrix A, the following are equivalent:

- *(a)* $\lim_{k\to\infty} A^k = 0$ *(i.e,* $\lim_{k\to\infty} \left\|A^k\right\| = 0$*).*
- *(b)* $\lim_{k\to\infty} A^k \overrightarrow{y} = 0,\ \forall \overrightarrow{y} \in \mathbb{C}^n$ *(i.e,* $\lim_{k\to\infty} \left\|A^k \overrightarrow{y}\right\| = 0$*).*
- *(c)* $\rho(A) < 1$*.*
- *(d) There exists a norm* $\|\cdot\|$ *such that* $\|A\| < 1$*.*

- Let $A = B$ in the above theorem. Then the equivalence of (b)(c)(d) gives the equivalence of (a)(b)(c) in the target theorem (I).

Convergence of the traditional iterative methods

Remark

Since

$$\vec{x}^{(k)} - \vec{x} = B^k \left(\vec{x}^{(0)} - \vec{x} \right),$$

then we have the following a priori error estimate:

$$\left\| \vec{x}^{(k)} - \vec{x} \right\| \leq \|B\|^k \left\| \vec{x}^{(0)} - \vec{x} \right\|.$$

Hence the convergence rate is $\|B\|$ *(or* $\rho(B)$*).*

Convergence of the traditional iterative methods

Theorem (II)

Consider the iterative method

$$\vec{x}^{(k+1)} = B\vec{x}^{(k)} + \vec{c}, \ k = 0, 1, 2, \cdots$$

Assume $\|B\| < 1$. *Then we have the following a posteriori error estimates:*

$$\left\|\vec{x}^{(k)} - \vec{x}\right\| \leq \frac{\|B\|}{1 - \|B\|}\left\|\vec{x}^{(k)} - \vec{x}^{(k-1)}\right\|.$$

and

$$\left\|\vec{x}^{(k)} - \vec{x}\right\| \leq \frac{\|B\|^k}{1 - \|B\|}\left\|\vec{x}^{(1)} - \vec{x}^{(0)}\right\|.$$

Convergence of the traditional iterative methods

Proof.

See page 146 of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Convergence of the SOR method

- Recall the SOR method:

$$\begin{aligned}\overrightarrow{x}^{(k+1)} &= \left(L+\frac{1}{\sigma}D\right)^{-1}\left(U+D-\frac{1}{\sigma}D\right)\overrightarrow{x}^{(k)} \\ &\quad +\left(L+\frac{1}{\sigma}D\right)^{-1}\overrightarrow{b},\ k=0,1,2,\cdots\end{aligned}$$

Convergence of the SOR method

Theorem (III)

If A is Hermitian positive definite and $0 < \sigma < 2$, then the SOR method converges for any initial vector.

Proof.

See pages 147-149 of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □

Convergence of the Jacobi and GS methods

- Recall the Jacobi method:

$$\overrightarrow{x}^{(k+1)} = -D^{-1}(L+U)\overrightarrow{x}^{(k)} + D^{-1}\overrightarrow{b}, \ k = 0, 1, 2, \cdots$$

Convergence of the Jacobi and GS methods

- Recall the Jacobi method:

$$\vec{x}^{(k+1)} = -D^{-1}(L+U)\vec{x}^{(k)} + D^{-1}\vec{b},\ k = 0, 1, 2, \cdots$$

- Rercall the GS method:

$$\vec{x}^{(k+1)} = -(L+D)^{-1}U\vec{x}^{(k)} + (L+D)^{-1}\vec{b},\ k = 0, 1, 2, \cdots$$

Convergence of the Jacobi and GS methods

Theorem (IV)

If A is a strictly or irreducible diagonally dominant, then the Jacobi method and GS method converge for any initial vector.

Proof.

See pages 150-153 of the textbook. (Independent study problem or take Math 6601 (Numerical Analysis)) □